Fooling Face Recognition Systems through Physical Adversarial Attack

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Abstract. Face recognition systems (FRS) are increasingly prevalent today worldwide. Their applications in the security sensitive fields such as verification and authorization, security, multimedia, and law enforcement are crucial and are ever on the rise. Most modern face recognition systems use convolutional neural networks (CNNs) as their backbone. This is because CNNs excel at feature extraction in images, and are also highly scalable and robust which makes them highly performant. However, these CNNs display some counterintuitive properties due to the linearities within the network. One such property enables an attacker to manipulate the predictions of the network by adding small, imperceptible, and appropriately calculated noise to the inputs of the neural networks. The fact that these kinds of attacks are stealthy and can be physically realizable, questions the reliability of systems using those networks. This paper proposes a physical attack technique against FRS using adversarial bandages/patches employing the iterative Fast Gradient Sign Method (FGSM). The results have been analysed using various sizes of adversarial patches. Furthermore, experiments have been performed considering both untargeted and targeted attacks with various target classes. The paper also proposes a way to select the best target class in targeted attacks, with justification behind such behaviour of targets. The physical realization of attacks through the generation of patches using FGSM has been performed using manually captured images. Finally, this paper completely explores the weaknesses and strengths of the model when being attacked by FGSM. This would motivate researchers to better defend their public CNNs and face recognition models, and improve their strategies for ethical and legal attacks of machine learning models.

Keywords: Adversarial bandage · Fast gradient sign method · Face recognition · Physical attack.

1 Introduction

Face recognition systems use softwares to identify or verify a person's identity by analyzing their face. These systems work in four steps: detecting the face in an image, capturing its keypoints, calculating features, and matching them against a database. This technology is used for security purposes like unlocking phones or in law enforcement for suspect identification. Convolutional neural networks (CNNs) excel at image recognition, and are vital for face recognition as they can effectively analyze facial features from an image. In short, CNNs power the system's ability to turn a face into a unique identifier. CNNs are a class of deep neural networks extensively used for image recognition and processing. They are designed to automatically and adaptively learn spatial hierarchies of features from input images. With their ability to handle large amounts of visual data and their hierarchical feature learning, CNNs have become integral in various fields including computer vision, medical image analysis, and autonomous driving. Despite the high accuracy, CNNs exhibit counterintuitive properties in their susceptibility to adversarial attacks which exploit vulnerabilities in CNNs by introducing subtle perturbations in the input to deceive them. Hence, despite their impressive performance in image recognition, CNNs can be easily fooled, leading to misclassification. These perturbations, often undetectable to human eyes, exploit the linearity and high-dimensional nature of CNNs. Additionally, CNNs tend to learn features that are not truly representative of the input, making them vulnerable to adversarial examples that manipulate these features. Moreover, CNNs lack robustness across different domains, as adversarial examples crafted for one network can often fool others with similar architectures. These properties highlight the fragility of CNNs and underscore the need for robust training techniques and defensive mechanisms to mitigate adversarial vulnerabilities.

1.1 Attack Techniques

There exist various methods to introduce these kinds of perturbations into an image. One of the most used is called Fast Gradient Sign Method (FGSM) [4]. It leverages the gradients of the loss function with respect to the input image where the loss function measures the deviation of the model's prediction from the ground-truth. By taking the sign of the gradients (positive or negative), FGSM identifies the direction in which each pixel value needs to be adjusted to maximize the loss function so that the image gets misclassified. These distortions are based on the model's own vulnerabilities, and push the image towards a classification the attacker desires. These perturbations are often imperceptible to the human eye, but significant enough to fool the model. Due to its simplicity and effectiveness, it is the most used attack method to explore the weaknesses of CNNs, so as to develop proper defence.

However, FGSM adds perturbations to the whole of the image. This cannot help in the physical realization of the method to fool a model as it is not feasible to add perturbations to the whole of the input to an object-, face- or scene-recognition model. Thus, we look into the possibilities of restricting the noise added to a particular region or within a boundary. Furthermore, we need to perturb some object that sits on the face instead of adding perturbations directly on the face. Then, it would be more feasible to physically realise the attack.

(a) (b) (c)

Fig. 1: Adversarial patch generation result: (a) Original image correctly classified as Tony Blair, and (b) adversarial patch causes it to be misclassified as Donald Rumsfeld. (c) Image of Donald Rumsfield. Image courtesy: LFW dataset [7].

1.2 Adversarial patch

Adversarial patch, first proposed by Brown et al. [1] is a small, carefully designed image or pattern that can be placed in a scene to cause a model to misclassify the entire scene. Unlike digital adversarial examples, which are pixel-level perturbations, the patch is a physical or digital overlay that can be used in real-world settings. Moreover, they are transferable [14], thus posing a significant concern in areas where machine learning systems are deployed in real-world applications, such as security and autonomous vehicles, and emphasize the need for robust defenses against potential attacks.

There are many possible objects that can serve as the *medium* for adversarial patches. However, to remain inconspicuous, it is best to use the ones that do not draw much attention, such as glasses, bandages or tattoos. These objects can be modified according to the determined adversarial perturbation. The aim then is to make it possible to misidentify a person by using images with these alterations. For its practicality and inconspicuousness, this paper selects bandages as the medium for introducing perturbation. Henceforth, we shall use the terms adversarial *patch* and *bandage* interchangeably. Figure 1 shows an image of Tony Blair being misclassified as Donald Rumsfeld upon attesting the adversarial bandage.

FGSM can be employed to execute two types of attacks: targeted and untargeted. Untargeted attacks are designed to cause the model to misclassify an image, regardless of the specific incorrect category. In this scenario, FGSM adjusts the gradients to maximize the total loss, thereby displacing the image from its original class in any direction. This often leads to classifications that deviate significantly from the intended class. Conversely, targeted attacks aim for a specific class as an outcome other than the true one. In this case, FGSM modifies the gradient calculations to direct the image towards the desired target class, necessitating some knowledge about the target category and the model's decision boundaries. These attack methods can also be applied to face recognition systems. Untargeted attacks can be used to mask one's identity by causing the model to recognize someone as a different individual, which can be advantageous for covert operations. Targeted attacks, however, can enable impersonation by making the model classify the attacker as a specific person, potentially undermining verification and authentication processes. This paper will investigate both attack types to evaluate their effectiveness.

Several factors can influence the success of FGSM-based attacks, such as the size and placement of the bandage or variations in facial features. This paper will examine these factors and their impact on the effectiveness of the attacks.

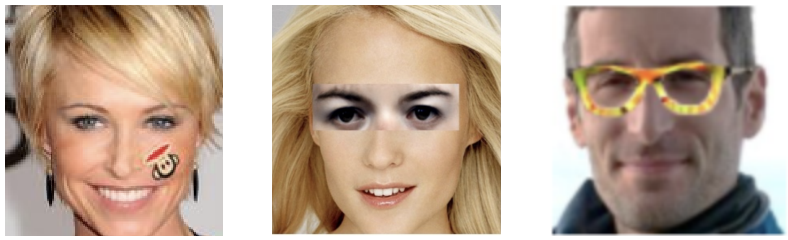
1.3 Novel Contributions

- Physical Implementation of Attacks: In addition to analyzing results from digital attacks, this paper explores the practical implementation of these attacks by physically applying adversarial patches to evaluate their effectiveness in real-world scenarios.
- Real-world Testing: Feasibility of using a perturbed bandage as an adversarial patch is investigated by printing and testing it against a face recognition model, assessing the physical realizability of the proposed attack method.
- Model Vulnerability Analysis: The paper provides insights into the vulnerabilities of face recognition models, contributing to a deeper understanding of their weaknesses, and informing strategies for improving defenses against convolutional neural networks (CNNs).
- Enhancement of Attack Strategies: Proposed methods for increasing attack success rates are discussed, offering potential improvements for executing effective and ethical adversarial attacks.
- Advancement of Stealthy Adversarial Attacks: The research advances the field of physical adversarial attacks by exploring the use of inconspicuous items, such as bandages, to achieve stealthy adversarial effects.

The paper is structured as follows. Section 2 discusses the related work on adversarial patch based attacks. Section 3 details the methodology while Section 4 presents the details of the experimentation followed by the results and discussion, in addition to results on physical realization of targeted and untargeted attacks. The paper then concludes in Section 5.

2 Related Work

The perplexing characteristics of deep neural networks have been under investigation since 2014, notably by Goodfellow et al. [11], who explored these behaviors, and proposed potential explanations for their counterintuitive nature. Carlini et al. [2] further examined how biases in deep neural networks can lead to unreliable and adverse outcomes, making them problematic for security-critical applications. Recognizing that these counterintuitive properties can cause inaccuracies in network predictions, research has shifted towards understanding and leveraging these behaviors. Goodfellow et al. [11] highlighted the vulnerability of



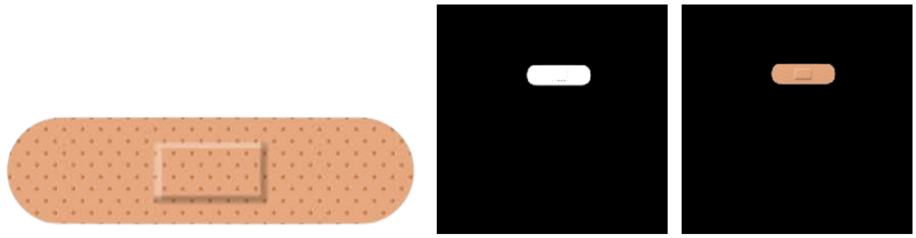
(a) (b) (c)

Fig. 2: Existing adversarial patch based attacks on: (a) face detection using stickers [13], and (b,c) face recognition: using adversarial patch overlaid on eyes [14] and spectacle frames [9], respectively.

convolutional neural networks (CNNs) to adversarial examples, and introduced the Fast Gradient Sign Method (FGSM) [4] as an effective technique for generating such examples using model inputs. Their findings indicate that adversarial examples can be so similar to legitimate ones that distinguishing between them visually can be challenging, demonstrating that even minor, imperceptible perturbations can mislead the model. Unlike traditional adversarial examples that alter individual pixels, patch-based attacks exploit the attention mechanism of object detectors, diverting focus to the patch instead of the actual objects [5]. These patches can be physically printed or displayed, posing a tangible threat to computer vision systems. Hu et al. [6] proposed crafting apparel with adversarial texture to fool person detection models while work in [3] fools the aerial object detectors.

In literature, similar attacks on face detection [13], [15] and recognition [14], [9], [10], [8] have been reported. Wei et al. [13] investigated using small, visually subtle stickers as shown in Fig. 2(a) that when strategically placed in a scene, fooled face detectors. Moreover, researchers demonstrated the practical application of adversarial patterns in fooling face recognition models. For instance, Xiao et al. [14] employed generative models to create adversarial patches as shown in Fig. 2(b) however, at the cost of obstructing vision since they cover the entire ocular region. Sharif et al. [9] used a 3D-printed eyeglass frame with the adversarial pattern worn by the attacker (Fig. 2(c)) to alter the system's performance. Authors proposed using multiple [10] or large grayscale stickers [8] on the face, but these are likely to be detected at any biometric deployment. A more comprehensive survey on attacks on face recognition can be found in [12].

Previous studies indicate that the areas of the face around the eyes and forehead play a significant role in face recognition. Therefore, applying an adversarial patch on these regions is optimal for carrying out a successful attack. While there have been attempts to use accessories such as spectacle frames [9] or emoji stickers [13] with adversarial perturbations, they can appear conspicuous and raise suspicion. Therefore, we have selected a bandage as the accessory owing to its potential for stealthiness.



(a) (b) (c)

Fig. 3: (a) The bandage image used in experiments, (b) the mask m indicating the location on the face for attesting the bandage, and (c) the patch template p.

3 Methodology

The objective of the attacker is to trick a face recognition model by applying a perturbed bandage on the face. This study will analyze how various factors, such as bandage size and facial features, impact the effectiveness of these attacks. Additionally, this work aims to identify optimal target classes, and propose alternative patch sizes and shapes to enhance attack success rates.

3.1 Prior Knowledge

To execute the attack described in this paper, the attacker requires access to the trained model used for image classification, as well as several input images. Consequently, this attack is categorized as a white-box attack, where the adversary has complete knowledge of the classifier, including details such as the input dimensions, the architecture, and the parameters of the feature extractor.

Details of the Image and Bandage: All input images are resized to 250×250 pixels. The bandage image used in experiments is shown in Fig. 3(a). The standard size for superimposing the bandage is 80×60 pixels, though its size is adjusted to study the effects of different dimensions. The bandage is positioned on the forehead, near the eyebrows, as this area is considered crucial for distinguishing facial features. This location for attaching the bandage is marked using a binary mask m (Fig. 3(b)). The bandage is overlaid onto an image x, and the patched image $\hat{x}$ is obtained by adding the patch template p (Fig. 3(c)) to the product of the mask complement and the original image as:

$$\hat{x} = (1 - m) \odot x + p \tag{1}$$

3.2 Algorithm

Iterative FGSM The Fast Gradient Sign Method (FGSM) is a popular and efficient technique in the field of adversarial machine learning. Since FGSM is a white-box attack, it necessitates an understanding of the parameters and design

Algorithm 1: Universal Targeted Adversarial Bandage Generation

Input: Image $x \in X$; target image t; patch template p_0; mask m; $epoch = 20$; $max\_iter = 500$; step size $\epsilon = 0.3$; $model$ = CNN; Classifier cls

Output: adversarial patch p

Initialize patch $p = p_0$ as shown in Fig. 3(c)
$t\_embed = model(t)$
for i = *1 to epoch* **do**
  for *each x in X* **do**
    for i = *1 to max_iter* **do**
      $\hat{x} = (1 - m) \odot x + p$
      $\hat{x}\_embed = model(\hat{x})$
      if $cls(t\_embed) = cls(\hat{x}\_embed)$ **then**
        break
      $L(\theta, \hat{x}, y_t)$ = cross entropy loss with true label y_t of target image t
      $\nabla L = grad(L, \hat{x})$
      $\nabla L = \nabla L \odot m$
      $p_{ae} = p + \epsilon \times sign(\nabla L)$
      $p = clip(p_{ae}, 0, 1)$
    end
  end
end
return p

of the model. The modus operandi is to use the gradient of the loss with respect to the input to determine the amount of noise/perturbation to be added in order to disrupt the input data. FGSM's simplicity comes from how well it generates adversarial samples at low computing cost. Let x be the input data, y the true label, and $L(\theta, x, y)$ the loss function of the model with parameters θ. Further, the adversarial example is generated as follows. The perturbation δ is calculated by taking the sign of the gradient as:

$$\delta = \epsilon \times sign(\nabla_x L(\theta, x, y)) \ , \tag{2}$$

where ϵ is the constant controlling the magnitude of the perturbation. Finally, the adversarial example x_{ae} is created by adding the perturbation to the original input as:

$$x_{ae} = x + \delta \, . \tag{3}$$

Iterative FGSM is a modified version of FGSM where the noise is calculated, and added using FGSM technique iteratively until the goal is reached, i.e., the image is misclassified. This technique can be used where just one pass of FGSM is not enough to confuse the model. It gives us more control over the amount of noise added. Usually, this also leads to higher success rates at the cost of a few more iterations. The methods for generating the adversarial bandage/patch using targeted and untargeted attacks are described next.

Targeted and Untargeted Attack: The objective of a targeted attack described in Algorithm 1 is to introduce adversarial noise into the image x so that

Algorithm 2: Universal Untargeted Adversarial Bandage Generation

Input: Image $x \in X$; patch template p_0; mask m; $epoch = 20$; $max\_iter = 500$; step size $\epsilon = 0.3$; $model$ = CNN; Classifier cls

Output: adversarial patch p

```
Initialize patch p = p_0 as shown in Fig. 3(c)
for i = 1 to epoch do
    for each x in X do
        x_embed = model(x)
        for i = 1 to max_iter do
            x̂ = (1 − m) ⊙ x + p
            x̂_embed = model(x̂)
            if cls(x_embed) ≠ cls(x̂_embed) then
                break
            L(θ, x̂, y) = −1 × cross entropy loss with true label y of image x
            ∇L = grad(L, x̂)
            ∇L = ∇L ⊙ m
            p_ae = p + ϵ × sign(∇L)
            p = clip(p_ae, 0, 1)
        end
    end
end
return p
```

it is classified to not just any other class but only to the target class y_t. To achieve this, FGSM is applied iteratively to the patch region on the patched image $\hat{x}$ until it is classified as the target class. In each iteration, the noise is computed using categorical cross entropy loss $L(\theta, \hat{x}, y_t)$, and the same is used to update the patch template p as:

$$\nabla L = grad(L, \hat{x})\,, \quad \nabla L = \nabla L \odot m\,, \quad p_{ae} = p + \epsilon \times sign(\nabla L) \tag{4}$$

The loss here is designed in a way that minimizing the loss shifts the prediction towards the target class.

Similarly, the aim of the untargeted attack is to introduce adversarial perturbation to the image x in a way that causes it to be classified into any class other than its original class. Thus, the approach here is to apply FGSM iteratively until the patched image $\hat{x}$ is classified into a different class. Different from the targeted attack, here the noise is determined using the negative categorical cross entropy loss $L(\theta, \hat{x}, y)$ computed using the true label y of the input $\hat{x}$. Thus reducing the loss results in the prediction deviating from the original class. The pseudocode is provided in Algorithm 2.

(a) Ariel Sharon (b) Donald Rumsfeld (c) John Ashcroft

(d) Junichiro (e) Serena Williams (f) Tony Blair

Fig. 4: Images of subjects from LFW Dataset used in this study

4 Experiments and Results

4.1 Dataset

The Labelled Faces in the Wild (LFW) dataset, a widely used benchmark for evaluating the performance of face verification algorithms is utilized. It consists of a collection of face images captured "in the wild" under uncontrolled conditions with variations in pose, lighting, expression and background clutter. This makes it a valuable resource for researchers developing algorithms that can accurately recognize faces in real-world scenarios as shown in Fig. 4. Though the dataset consists of more than 1000 classes, it can be time and resource consuming to experiment on all the classes. Moreover, only 12 classes in the dataset have more than 40 images each. Thus, for experimental convenience and better results, 6 classes from the 12 most populous classes are selected for experimenting.

4.2 Model Training

Details of the classifier: For the experimentation, as already mentioned earlier, 6 classes out of the most populated classes were used. Using a benchmark face recognition model like Facenet, Arcface etc. would overfit the data into the model as those networks are extremely complex when compared to the limited input that is being used. Thus, a custom CNN was used to train the model. It consists of two convolution layers (kernel size 3×3 and 32 filters each) with ReLU activation followed by a 2×2 maxpooling layer, two fully connected (FC)

Table 1: Original test accuracies (in %) for the six classes (Row 1), and accuracies (in %) after untargeted attack (Row 2) and targeted attacks (Rows 3-8)

Test Accuracies	Ariel Sharon	Donald Rumsfeld	John Ashcroft	Junichiro	Serena Williams	Tony Blair
Original test accuracy	80	60	100	100	85	70
Untargeted Attack	0	10	0	100	35	0
Target class 1 (Ariel S)	100	10	0	10	80	40
Target class 2 (Donald R)	0	100	0	0	0	0
Target class 3 (John A)	10	0	100	0	60	10
Target class 4 (Junichiro)	0	10	0	100	20	0
Target class 5 (Serena W)	0	0	0	80	100	40
Target class 6 (Tony B)	0	0	50	0	30	100

(a) (b) (c) (d)

Fig. 5: Result of (a,b) targeted and (c,d) untargeted attack: (a) Adversarial bandaged image of Donald Rumsfeld (generated using Algorithm 1) is misclassified as Tony Blair (target class) and (b) Tony Blair image is correctly classified. (c,d) Adversarial bandaged images of Donald Rumsfeld and Tony Blair (generated using Algorithm 2) are misclassified.

layers (128 and 6 neurons, respectively), and finally a softmax layer for calculating the cross entropy loss. Adam optimizer was used during training. The model was trained on the classes for 15 epochs. The class-wise accuracies achieved are shown in Table 1 (Row 1).

4.3 Accuracies After Attack

Using Algorithms 1 and 2, both targeted and untargeted attacks have been performed on the data, and the corresponding attack results are demonstrated in Figs. 5(a,b) and 5(c,d), respectively. Table 1 shows the test accuracies of all the classes after untargeted (Row 2) and targeted attacks (Rows 3-8). Untargeted attack misclassifies a decent proportion of images in the dataset as evident by the reduced accuracy of majority of the classes. For instance, in case of John Ashcroft, the accuracy dropped from 100% to 0% implying that when adversarial examples generated from samples of J. Ashcroft are given as input to the model, all of them get misclassified into another class. The targeted attack also has performed as expected in most cases. The exception in both the cases is the class Serena Williams which the algorithm struggled to misclassify. This could be seen as one such example where the sheer facial features are so distant that the attack is not so easy to perform (Serena Williams is a black woman where all others are white/Asian men). Other little discrepancies in the accuracies can also be accounted for by distant facial features or inaccurate learning of the model. It may be noted that not all target classes are equally vulnerable. Among the considered classes, Target class 2 (Donald Rumsfeld) is the most easy target class as the adversarial examples generated from all other classes are misclassified into it. On the other hand, classes 1 and 5 seem to be fairly difficult to target.

4.4 Model Vulnerability Analysis

Choice of target class: A well-performing model is typically expected to classify inputs with high confidence, indicating robustness where minor variations in the input do not significantly impact its predictions thereby, resulting in high prediction scores for the correct class and very low scores for all other classes. However, this characteristic can pose challenges while using gradient-based methods such as IFGSM to introduce adversarial perturbation since it adds adversarial noise in very small increments, which alters prediction scores by a very small value. As a result, generating an adversarial example for such models becomes challenging.

Table 2 shows the prediction scores computed for five samples *Sample-x* each belonging to a different class, Class x. It can be observed that for the correct classes, prediction scores are as high as 0.99, while for the incorrect classes, prediction scores range from 10^{-4} to 10^{-7}. In such a scenario, selecting these incorrect classes as targets generally results in poor attack success rates. For instance, when a sample belonging to Class 2 is given as an input, the softmax probability corresponding to Class 4 is merely 2.7e−7. Consequently, if Class 4 is chosen as the target class, it is highly unlikely that images of Class 2 will be misclassified as those of Class 4. This behavior can arise due to several factors such as well-separated decision boundaries established during training or dataset containing easily distinguishable classes perhaps owing to significant differences in facial features between individuals. For example, a man with a round face may be misclassified as another man with a round face, but would be less likely to be misclassified as a woman of a different race. Such differences are beyond control at the input level. Therefore, target classes must be chosen while keeping the characteristics of the original class in mind.

Effect of bandage size: In our experiments, the image size was fixed to 250×250 pixels, and the bandage size was fixed to 80×60 pixels, covering approximately 7.68% of the image area. This size was chosen to ensure that the bandage neither appears too small nor too large on the face. Appropriate bandage size is important, particularly when the model classifies with a high confidence score. Further, to analyse the potential effect of the bandage size on the attack success rate, we considered three different bandage sizes, 4.88%, 7.68% and 14.50% of the image size as shown in Fig. 6. As can be observed, the

Table 2: Class-wise prediction scores by the model

Predictions	Class 1	Class 2	Class 3	Class 4	Class 5
Sample-1	9.8e−1	3.8e−3	3.7e−5	3.0e−4	9.5e−3
Sample-2	2.0e−4	9.9e−1	2.8e−3	2.7e−7	6.3e−5
Sample-3	7.4e−7	1.2e−4	9.8e−1	2.0e−2	2.4e−4
Sample-4	8.7e−7	1.5e−4	6.8e−3	9.9e−1	2.9e−4
Sample-5	1.8e−2	3.6e−7	5.4e−7	4.8e−5	9.8e−1

(a) (b) (c)

Fig. 6: Effect of size of bandage in terms of its area in proportion to the image: (a) 4.88%, (b) 7.68% and (c) 14.50%. The extreme, moderate and minimal distortions visible on the bandages in (a), (b) and (c), respectively, indicate that high, moderate and low amount of noise were added to these bandages. For privacy preservation, parts of the face are hidden.

smallest patch exhibits more deviation from the original bandage's appearance with a considerable distortion in color, and requires more number of iterations of IFGSM in order to be effective as an attack while larger patches require lesser perturbation and fewer iterations to achieve misclassification, potentially leading to a higher overall rate of misclassification. However, using bandages of abnormally large sizes presents challenges in terms of deception since they may appear unrealistic and raise suspicion. On the other hand, smaller bandages can be less effective and exhibit unusual color distortions.

4.5 Physical Realisation

Physical realization of the adversarial attack can be challenging due to several factors such as ambient lighting, camera settings, blur and other noises introduced in the captured images, rendering the adversarial attack ineffective. The objective of this paper is to assess the practicality of physically implementing an adversarial patch using the I-FGSM algorithm. To this end, a dataset of face images was manually collected, comprising 40 images of 4 individuals. Patch size was selected considering the tradeoff between the resemblance with a traditional bandage and magnitude of adversarial perturbation as discussed in Section 4.4. Figure 7 shows a subject attesting the bandage on the forehead in order to launch an adversarial attack. The data was divided into training, validation, and test sets with suitable proportions. The previously described model was trained on this dataset for 10 epochs, and achieved perfect classification across all classes with 100% accuracy. Subsequently, both targeted and untargeted attacks were carried out using this dataset and the trained model, as discussed further.

(a) (b)

Fig. 7: Example of attesting a physical bandage. (a) Image of ID_3 without bandage; (b) misclassified as ID_4 with the adversarial bandage.

Targeted and Untargeted Attack: Table 3 shows that targeted attacks are successful for all classes. Further, we tried to identify the optimal combination of source class (original class of input) and target class such that it results in a more successful targeted attack, leading to a more effective attack strategy. Ideally, a class having images bearing high facial similarity with the source class' images can be a suitable target class. To determine the suitable pair(s), we assume two scenarios: the source class is fixed and the target class is determined or the source class is determined by fixing the target class. Both the scenarios correspond to impersonation, while in the first scenario, the attacker intends to get authenticated as any enrolled user. On the other hand, in the second scenario, any enrolled accomplice may masquerade as belonging to a specific target class. To determine the best source-target class pairs suitable for both the scenarios, clustering of the prediction vectors (vector comprising prediction scores of all classes) obtained from all the subjects can be performed with number of clusters less than the number of classes. The best source-target class pair is the one having the minimum distance between the prediction vectors within a cluster.

For example, Fig. 8 shows results of K-means clustering on prediction vectors of the samples of the 4 classes considered here, and k is set to 3. For better visualization, two principal components are shown, with violet colored cluster containing two data points suggesting that selecting the classes within this cluster, one as a source and the other as a target is likely to yield the most successful targeted attack. Moreover, this information can also be utilized to develop more robust adversarial defense, specially designed to defend against easier target classes. Untargeted attack was also attempted on the model. Since the clusters are far enough, the results were near ideal as shown in Table 3 (Row 2).

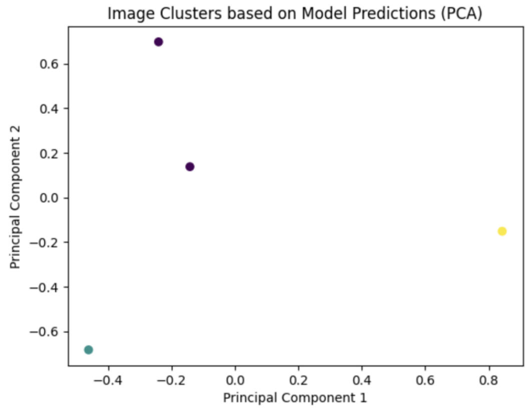


Fig. 8: Depiction of the three clusters formed indicated in violet, green and yellow.

Table 3: Original test accuracies (in %) for the four classes (Row 1), and accuracies (%) after untargeted attack (Row 2) and targeted attacks (Rows 3-6)

Test Accuracies	ID_1	ID_2	ID_3	ID_4
Original test accuracy	100.0	100.0	100.0	100.0
Untargeted Attack	12.5	0	39.0	0
Target class ID_1	100.0	0	0	0
Target class ID_2	0	100.0	0	12.5
Target class ID_3	12.5	0	100.0	0
Target class ID_4	0	100.0	0	100.0

5 Conclusion

Face recognition systems are predominantly employed in security-sensitive applications, such as authorization, necessitating high reliability and robustness. However, deep neural networks, which form the backbone of these systems, are known to exhibit counterintuitive behaviors that can be exploited to deceive the network. Understanding these vulnerabilities is crucial for developing defenses against potential attacks that could alter the network's outputs. This paper investigates the feasibility of deceiving a convolutional neural network using Iterative FGSM. Both targeted and untargeted attacks were conducted to observe the model's behavior. The best classes and results were selected from numerous trials involving various classes and parameters. The study found that some classes are more effective targets in targeted attacks than others. To identify the optimal source and target class pairs, clustering was performed on the dataset's prediction values. Additionally, the paper explores the physical implementation of adversarial attacks using a bandage. A dataset of manually captured images was created for this purpose, and physical bandages were produced and tested. The model's predictions on these images aligned with expectations. This research addresses key considerations for both executing and defending against FGSM attacks, providing insights to enhance model robustness. Future work could extend these findings to larger datasets and different types of models to evaluate the generalizability of the results.

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References

1. Brown, T.B., Mané, D., Roy, A., Abadi, M., Gilmer, J.: Adversarial patch. arXiv preprint arXiv:1712.09665 (2017)
2. Carlini, N., Wagner, D.: Towards evaluating the robustness of neural networks. In: IEEE Symposium on Security and Privacy (SP). pp. 39–57 (2017)
3. Du, A., Chen, B., Chin, T.J., Law, Y.W., Sasdelli, M., Rajasegaran, R., Campbell, D.: Physical adversarial attacks on an aerial imagery object detector. In: Proceedings of the IEEE/CVF WACV. pp. 1796–1806 (2022)
4. Goodfellow, I.J., Shlens, J., Szegedy, C.: Explaining and harnessing adversarial examples. arXiv preprint 1412.6572 (2014)
5. Hu, Y.C.T., Kung, B.H., Tan, D.S., Chen, J.C., Hua, K.L., Cheng, W.H.: Naturalistic physical adversarial patch for object detectors. In: Proceedings of the IEEE/CVF International Conference on Computer Vision. pp. 7848–7857 (2021)
6. Hu, Z., Huang, S., Zhu, X., Sun, F., Zhang, B., Hu, X.: Adversarial texture for fooling person detectors in the physical world. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. pp. 13307–13316 (2022)
7. Huang, G.B., Ramesh, M., Berg, T., Learned-Miller, E.: Labeled faces in the wild: A database for studying face recognition in unconstrained environments. Tech. Rep. 07-49, University of Massachusetts, Amherst (October 2007)
8. Pautov, M., Melnikov, G., Kaziakhmedov, E., Kireev, K., Petiushko, A.: On adversarial patches: real-world attack on arcface-100 face recognition system. In: (SIBIRCON). pp. 0391–0396. IEEE (2019)
9. Sharif, M., Bhagavatula, S., Bauer, L., Reiter, M.K.: Accessorize to a crime: Real and stealthy attacks on state-of-the-art face recognition. In: Proc. ACM Sigsac Conference on Computer and Communications Security. pp. 1528–1540 (2016)
10. Shen, M., Yu, H., Zhu, L., Xu, K., Li, Q., Hu, J.: Effective and robust physical-world attacks on deep learning face recognition systems. IEEE Transactions on Information Forensics and Security **16**, 4063–4077 (2021)
11. Szegedy, C., Zaremba, W., Sutskever, I., Bruna, J., Erhan, D., Goodfellow, I., Fergus, R.: Intriguing properties of neural networks. arXiv preprint 1312.6199 (2013)
12. Vakhshiteh, F., Nickabadi, A., Ramachandra, R.: Adversarial attacks against face recognition: A comprehensive study. IEEE Access **9**, 92735–92756 (2021)
13. Wei, X., Guo, Y., Yu, J.: Adversarial sticker: A stealthy attack method in the physical world. IEEE TPAMI **45**(3), 2711–2725 (2022)
14. Xiao, Z., Gao, X., Fu, C., Dong, Y., Gao, W., Zhang, X., Zhou, J., Zhu, J.: Improving transferability of adversarial patches on face recognition with generative models. In: Proceedings of the IEEE/CVF CVPR. pp. 11845–11854 (2021)
15. Yang, X., Wei, F., Zhang, H., Zhu, J.: Design and interpretation of universal adversarial patches in face detection. In: ECCV. pp. 174–191. Springer (2020)